

4 A The mass slowed down at an increasing rate from $t = 0.25$ s to $t = 0.50$ s.

5 C $(\Sigma P_i = \Sigma P_f)_y \uparrow +$
 $0 = (h/\lambda_f) \sin 75^\circ - p \sin \alpha$
 $(6.63 \times 10^{-34} / 966.8 \times 10^{-12}) \sin 75^\circ - (8.362 \times 10^{-25}) \sin \alpha$
 $\sin \alpha = 0.7922$
 $\alpha = 52.4^\circ$